# Vishnu Unnikrishnan

Research Data Science @ Pfizer | Master's in Statistics | Cochin University of Science and Technology.
LinkedIn | Personal Webpage | vishnuunk@gmail.com | +91-9446190855

## Experience

**Research Intern, Biostatistician** **May 2026 – Ongoing**
*Pfizer India* *Chennai, India*
- Working with the Translational Biomarker Statistics Team within Pfizer Data Science & Analytics (DSA) on developing a novel methodology for high-dimensional feature selection for biomarker data. **Publication in progress**.
- Research work broadly lies in areas of Precision Medicine, Omics Biomarkers, High Dimensional Statistics and ML.

**Research Intern, Biostatistics** **May 2025 – July 2025**
*ICMR - National Institute of Epidemiology* *Chennai, India*
- Worked on Multi-state modeling of longitudinal Mucormycosis patient clinical progression in India using semi-Markov proportional hazards model. Built fully reproducible pipeline using `R {mstate, survival}, Excel and Stata.`

**Data Science Intern** **Dec. 2022 – Jan. 2023**
*Kerala State Economics and Statistics Department* *Kerala, India*
- Worked on 'Predictive modeling of Paddy Production and Yield' project for the Government of Kerala. Utilized Time Series models to forecast paddy yield (Winter Season 2023) based on 4 years of data (varieties, cultivation area, irrigation etc.). Delivered critical data-driven insights for government agricultural planning and policymaking.

## Projects

**Multi-Omics Prognostic Biomarker Identification (Master's Thesis)** | *R, Machine Learning, Survival Analysis*
- Preprocessed 57K multi-omics features for 439 LUAD patients, deployed Cox-SIS to reduce dimensionality to 288 targets
- Trained penalized Cox & ensemble survival models via 100-subsample Stability Selection to isolate robust biomarkers
- Derived a 9-gene consensus panel & validated risk stratification via cross-validated C-Index & log-rank tests

**Stacked Ensemble Prediction of Fatal Myocardial Infarction Complications** | *Python, Machine Learning, Data Science*
- Tackled class imbalance using SMOTE & extracted top 22 clinical predictors out of 111 via LASSO & RF-based RFE
- Trained LR, kNN, GBM, RF & XGBoost via 10-fold CV & stacked predictions via GLM & RF meta-learners
- Attained 0.87 Test Accuracy, 0.93 F1 & 0.81 AUC using Stacked RF, with 97.4% sensitivity in identifying fatal cases.

**Unsupervised Transcriptomic Subgroup Discovery in Leukemia** | *Python, Machine Learning*
- Filtered 22k microarray probes to 1k via variance & extracted transcriptomic embeddings using PCA, t-SNE & UMAP
- Clustered PCA space via K-Means, Agglomerative & GMM, attaining 0.85 NMI with BIC-optimized GMM (k=8)
- Identified 988 significant biomarker probes via ANOVA F-testing & Benjamini-Hochberg FDR correction

**Survival Analysis of Systolic Blood Pressure & Mortality** | *SAS, Survival Analysis*
- Modeled 22-year mortality on 2k Framingham cohort patients via multivariate Cox PH & AIC-driven stepwise selection
- Evaluated PH assumptions via Schoenfeld residuals & time-varying plots, resolving violations via age stratification
- Identified aHR of 1.014 ($p < 0.0001$), indicating a 1.4% mortality risk increase per systolic BP unit

**Financial Time Series Forecasting and Risk Analysis** | *Python, Time Series Analysis, Deep Learning*
- Engineered 10y TSLA/SPY/BND market data to build ARIMA, SARIMA & LSTM forecasts.

## Education

**Cochin University of Science and Technology, Kerala, India** **July 2024 – July 2026**
*Master of Science (M.Sc.), Statistics*

**St. Thomas College (Autonomous), Kerala, India** **Aug. 2020 – July 2023**
*Bachelor of Science (B.Sc.), Mathematics*

## Skills

**Technical:** Python, R, SAS, Excel, SQL, Power BI, Tableau, Github, Data Science, Machine Learning, Deep Learning
**Interpersonal:** Communication, Teamwork, Leadership, Problem-solving
**Courses/Certificates:** Machine Learning Specialization, R Developer, Statistician in R, Statistics for Clinical Research (SAS, SDTM, ADaM, TFLs) - IQVIA, SAS Programming, Design of Experiments, Data Scientist tech stacks

## Achievement / Leadership / Extracurricular

- **State First** (3x) - Maria Philip Memorial Debate Competition (2020,2021,2022). **State Finalist** - Ministry of Youth Affairs and Sports' National Youth Parliament (2023,2024). **Head/President** - Debate and Oratory Club, STC.
- Cleared **GATE ST** (2026), **GATE DA** (2025), IIT JAM MS (2024), **ISI** PGDSMA 2023 (AIR 43), UPSC NDA (2020)
- Qualified for **National Finals** - **Bitathon Data Science Hackathon** 2026, GIM Goa
- Conference Coordinator & Volunteer Lead - IISA Annual Conference 2024, ICSSA Conference 2026